INTRODUCTION
WHY MANAGING QUALITY IS ESSENTIAL IN PROJECT MANAGEMENT

WHY PROJECT QUALITY?

We all agree and accept that as an end user of a product or service we would like what we order, when we want it and on the basis of good value for money. In a competitive world of consumer choice, we also expect it to last. We understand the benefits of a market-driven economy and need for appropriate product and service quality. This is the domain of operations, services and supply chain management. And we define our expectations as 'quality is what customer expects as a lasting experience' (Basu, 2011a). However, in the field of project management the importance of quality is not so clear-cut. Project managers appear to accept the 'iron triangle of cost, budget and quality' (Atkinson, 1999) but focus more on 'on time and budget' delivery as the success factors. Quality in projects is often relegated to mere 'lip service' and to tick-box compliance.

Project managers also appreciate the risk of a project on the basis of its uniqueness, complexity and deliberate design but appear not to connect the outcome of risks with the root causes underpinned by the dimensions of project quality. As a consequence we find many examples (as described later) of projects which were delivered on time and within budget but failed to meet the expectations of end users in the longer run. We need to ask ourselves 'how diligent are we in terms of project processes to deliver project objectives'? This is the minimum requirement to meet the customer needs. Furthermore we should also investigate 'how good is our project management … as a vehicle for delivering the longer-term outcomes and benefits as required by the sponsors and end users'. This is an element of the additional requirements of sustainable quality leading to project excellence. The relationship between *project quality* and *project excellence* will be discussed in more detail later.

Let us have a look at the current thinking and practices around these questions. The extant project management literature (Turner, 1999; Atkinson, 1999: Meredith

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and Mantel, 2003; Morris and Hough, 1997) identifies and supports three criteria or objectives for assessing the success of a project known as the 'iron triangle 'of time, cost and quality. The first two objectives are relatively simple to define and measure (Morris and Hough, 1997). Project quality as the third objective or dimension of the iron triangle is more difficult to define and assess, although it has received some attention in the academic literature (Turner, 2002; Heisler, 1990). Turner (2002) is among the few authors who attempts to more clearly define project quality comprising two dimensions as product quality and process quality. The guidelines for project quality in the project management bodies of knowledge (Association of Project Management, 2007; Project Management Institute, 2008) also reflect procedures of design and process requirements. These definitions and guidelines appear to suffer from two important limitations, viz. a lack of clarity in the definition (Whitty and Schulz, 2005) and the exclusion of organisational learning practices (Kotnour, 2000).

The lack of clarity around quality is often the source of project disputes and there are plenty of examples in the business world, as illustrated in the box, documenting the link between inadequate attention to quality management and unsuccessful major projects.

CASE EXAMPLE THE MILLENNIUM DOME

'The Millennium Dome project was one of the most controversial public works projects ever undertaken' (National Audit Office, 2000). The National Audit Office report also stated that the New Millennium Experience Company experienced severe financial difficulties. The main cause of these difficulties was the failure to achieve the visitor numbers: other contributing factors included the quality of project delivery and the contents within the dome.

CASE EXAMPLE WEMBLEY STADIUM

'The company that built the new Wembley Stadium, which opened after years of delays and almost tripling its cost, is suing the engineering consultants behind the project for £253m, claiming that their services were unsatisfactory' (*The Observer*, 16 March 2008). A preliminary search of legal cases (British and Irish Legal Information Institute, http://www.bailii.org, accessed 26/11/08) indicated several instances (2512 hits) of litigations because of 'poor quality' in projects. For example, in the recent Wembley Stadium project, there were eight major litigations related to project quality and three of these litigations were related to the definitions of project quality. In the case between Multiplex Construction Ltd and Honeywell Control Ltd (both being the contractors of Wembley Stadium) the dispute was to resolve the statement in the contract, 'It will have extensive, high-quality corporate hospitality facilities and a state of the art communications system (installed by Honeywell).' (Neutral Citation Number: [2007] EWHC 447 (TCC), www.bailii.org, accessed 26/11/08).

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CASE EXAMPLE WEST COAST RAIL UPGRADE

The rail line between Glasgow and London was undergoing an £8.6bn upgrade from 2003. The modernisation of the West Coast Main Line will deliver the following enhancements:

- 125mph route capability for tilting trains delivering much faster journey times.
- Capacity for significantly more long-distance passenger and freight trains than today.
- Better and more resilient performance in travel time and safety measures.

The National Audit Office said it might not be able to cope with current levels of growth beyond 2015. The auditors' report on the West Coast Line warned that electronic signalling equipment might become obsolete significantly earlier than expected. The auditors were also concerned about the ineffective communications between key stakeholders (Government, Network Rail, Rail Track and Virgin Trains). To sustain train operations, the line's operator, Virgin Trains, was paid £590m more in subsidy in the period 2002–06 than envisaged in its franchise agreement, their report said. In January 2008, an over-run on work results in one of the worst delays yet. Network Rail is fined £14m.

CASE EXAMPLE HEATHROW TERMINAL 5

'Thousands of suitcases are being sent to Milan by British Airways to try to help clear a backlog of 19,000 bags at Heathrow's new Terminal 5' (*The Guardian*, April 2 2008). The conclusions of the House of Commons Transport Select Committee Report (2008) highlighted that insufficient communication between the owner of the new terminal, BAA, and its operator, British Airways, was a major factor in the ensuing complications with the baggage system and security searches. As it will be discussed later, communication could be an essential component of people-related quality.

The above examples of major project failures appear to focus on the quality of design, the quality of execution processes and the quality of communications between stakeholders. Many papers and studies in the 1990s (Belassi and Tukel, 1996; Tam, 1999; Kirby, 1996) highlighted project failures, but the problems still exist (MPA, 2003). Recent academic publications (Jamieson and Morris, 2008; Ling, Liu and Woo, 2008; Zou, Zhang and Wang, 2007; Abdelsalam and Gad, 2008) also suggest that causes of project failures include inadequate risk evaluation and quality management. These papers also highlight that there is a lack of clarity regarding the dimensions of project quality and its application with key stakeholders.

When you look at the domain of operations management you may observe some proven paths to follow. Operations management enjoys some success stories (along with the occasional failure) in terms of the application of quality-based operational excellence concepts such as total quality management, Six Sigma, lean and supply chain management (Oakland, 2003). The application of operational excellence concepts are now extended to non-manufacturing processes.

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> *Firms such as Motorola, General Electric ... successfully implemented Six Sigma. Motorola saved $15 billion in an 11 year period. General Electric saved $2 billion in 1999 alone ... Although Six Sigma initiatives have focused primarily on improving the performance of manufacturing processes, the concepts are widely applied in non-manufacturing, administrative and service functions. (Weinstein, Castellano, Petrick and Vokurka, 2008)*

Even though operational excellence concepts (such as Six Sigma) are often driven by the objective of cost-effectiveness the enablers of these concepts are rooted to the fundamentals of quality management (Oakland, 2003).

In the domain of operations management, the dimensions and definitions of quality have been identified by some authors (Garvin, 1984; Parasuraman, Zeithamel and Berry, 1984). The early leaders of total quality management (TQM) (Deming, 1986; Juran, 1989; Feigenbaum, 1983) emphasised the importance of people-related issues as a dimension of quality. On the other hand, the dimensions and definitions of quality appear to be wanting in publications related to project management. Project management standards (e.g. PMBOK, 2008; PRINCE2, 2009) focus primarily on processes in the project life cycle with some references to quality management systems. Turner (2002) appears to agree with Wild's (2002) dimensions of product quality and process quality. Kotnour (2000) points out the lack of clarity in the definition of project quality and the role of organisational learning in project management. There are publications regarding the success criteria and success factors of projects (Pinto and Slevin, 1988; Grude, Turner and Wateridge, 1996) but their implications in the dimensions of project quality are not clear. The application of excellence models in projects appears to be limited (Westerveld, 2003). Unlike operations management, the tools and concepts of operations excellence (Basu, 2008), such as lean and Six Sigma, are rarely being applied in project management (Pinch, 2005).

As summarised in Table I.1, the publications for the manufacturing and service sectors are more focused on the definitions of quality with distinctive dimensions and also encompass operational excellence concepts. The publications for the project management sector include success criteria and success factors, but these in turn appear to show a gap with their link with the dimensions of project quality. There is also an apparent gap between the application operational excellence concepts in project management.

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Table I.1 Classification of publications on quality and excellence

Publications	Dimensions and definitions of quality Product process people	Success criteria and success factors	Excellence models	Operational excellence concepts SCM/ TQM/6Sigma	Application sector
Gravin (1984)	x*				Manufacturing
Parasuraman, Zeithamel and Berry (1984)	x				Service
Wild (2002)	x x x				Manufacturing
Deming (1986)	x x			x	Manufacturing
Juran (1989)	x x			x	Manufacturing
Feigenbaum (1983)	x x			x	Manufacturing
Whitty and Sculz (2005)	x				Projects
PRINCE2 (2009)	x				Projects
Turner (2002) Kotnour (2000)	x x x				Projects Projects
Pinto and Slevin (1988)		x			Projects
Grude, Turner and Wateridge (1996)		x			Projects
Westerveld (2003)			x		Projects
This book	**x x x**	**x**	**x**	**x x**	**Projects**

* x denotes the presence of the topic in the publication.

On the basis of this summary there appears to be a knowledge gap in project management related to the benefits achieved by quality management when compared to the manufacturing and service operations. This book investigates the impact of all aspects of quality management in project management (see Table 1.1). The aim of the book is to contribute to this knowledge gap in project management and take the status of subject matters a step forward.

Furthermore, this preliminary review also indicates that there should be a clearer definition of project quality to establish some key dimensions. If you cannot define, you cannot measure, and if you cannot measure, you cannot control, assure or improve. First, what are the dimensions of project quality? Is it meeting specifications, timescales, budgets, achieving its purpose, producing happy project teams and stakeholders? Or does project quality constitute a combination of all these?

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Secondly, what tools or models can be effectively used in the quest for achieving and sustaining project excellence? What factors and processes are important in assessing the maturity and performance of projects?

Thirdly, how can the successes of operational excellence concepts, such as supply chain management, lean thinking and Six Sigma, be gainfully deployed in enhancing project quality and excellence?

THE AIMS AND PURPOSE OF THIS BOOK

This book aims to examine how quality should be defined and measured in the management of projects, and how the best practices of operations management can be applied to improve the likelihood of achieving project excellence. The investigation focuses initially on defining the dimensions of quality in project management and identifying sources of measurement for project excellence. Although there are a number of tools that measure excellence such as the balanced scorecard (Kaplan and Norton, 2004) or the EFQM model (2003) in operations management, it could be argued that the jury is still out (Association of Project Management, 2007) as how to apply these tools to achieve the level of longer-lasting performance in projects. An in-depth analysis will critically evaluate the assertions behind why it is important to include the dimensions of quality, excellence models and operational excellence methodologies in delivering sustainable outcomes of projects. The subsequent data collection phase of the field research will be validated by a causal research model, and further examined by case studies. These case studies are expected to open opportunities for further research in applying the best practices in different types of projects and programmes in construction, information technology and change management.

The purpose of this book is to go beyond the traditional measures of risk assessment, cost and time management in a project. It has been argued by Westerveld (2003) and Wateridge (2002) that although there have been significant publications on both project success criteria and critical success factors for projects, there has not been adequate research to define a concept that can link the two. There are also signs of visible efforts by IPMA (International Project Management Association) (e.g. the 'project excellence' model) and PMI (Project Management Institute) (e.g. the 'OPM3' model) to assess a project organisation's maturity and capability leading to project excellence. However, there appear to be no practically grounded and academically robust links between project excellence processes and project quality dimensions.

ISO 10006, in spite of its title – *Quality management - guidelines to quality in project management* – appears to provide the opposite effect in managing quality in projects (Stanleigh, 2007). If more attention were given to the items of written

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procedures at the expense of other critical people-related elements, then the project could become unsuccessful, although in theory it may remain compliant with standards. Other documents of project methodology also appear to offer little more than mere lip service regarding quality management. For example, BS6079 (2002) does not cover quality (other than project control) and PRINCE2 (2009) methods provide limited guidelines for quality planning and quality control within the process 'planning quality' module and reflect internally focused design and process requirements.

In summary, you may understand the role of quality in project management, but this alone is not sufficient. There is a far greater need to appreciate how one can define and measure the dimensions of quality – especially the people-related sustainable criteria – and then apply them towards the goal of achieving project excellence.

Against this backdrop, this book will fill the gaps in measuring project quality and project excellence.

Based on the initial review and apparent gaps identified, the key research question in this book has been formulated as: *How can project quality enable major projects deliver sustainable outcomes to key stakeholders leading to project excellence?* For the purpose of a focused enquiry in the literature, this overall research question has been broken into four supplementary areas:

1. What are the current processes of measuring the dimensions of quality in various stages of the project life cycle and what are the sustainable criteria?
2. What are the current models and approaches of assessing project excellence and what are the links between project success criteria and critical success factors?
3. Can a holistic model measure the dimensions of quality of a project and also assess the overall effectiveness of a project and project organisation?
4. How practically are the operational excellence approaches likely to be used in project management and how do we relate the outputs of such approaches to project excellence?

MAJOR PROJECTS

The case examples presented earlier in the chapter underline the importance of project quality more significantly attributed to major infrastructure projects. The infrastructure includes highways, railways, ports, tunnels, bridges, power plants, mass transits, municipal facilities or a combination of similar public facilities with the primary purpose of serving public needs, providing social services and promoting private economic activities (Algarni, Adriti and Polat, 2007). According to the recent UK Infrastructure Report (BMI, 2008):

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> *As one of the most robust economies in the world, the UK boasts of a large and highly developed construction industry, currently valued at over £80 billion. The UK construction industry is characterised by a small number of large players that have access to huge capital reserves and world-class technology, enabling them to undertake massive infrastructure development projects. Recent major infrastructure projects include the planned development of the Crossrail in London at an estimated cost of £18 billion, the £5.8 billion development of the Channel Tunnel Rail Link, and the £4.3 billion construction of Terminal 5 at Heathrow Airport.*

In comparison to other major undertakings, the major infrastructure projects in the UK may have to face additional technical and management challenges, such as wide range of stakeholders including consumers, multifunctional features in building, roads, rails, tunnels and systems and above all, a long public review period before it can take off. The demand on project quality management is also likely to be more challenging in keeping with the complexities of a major infrastructure project.

With the above rationale, the research, particularly the case studies, will focus primarily on the issues, challenges and opportunities in project quality and excellence in major infrastructure projects in the UK. The infrastructure initiatives also cover systems to deliver sustainable logistics service requirements of a community and country as part of local and global requirements.

It is intended that conclusions and recommendations from this book will provide a foundation to extend the definition and implementation of project quality and excellence to all types of projects in all parts of the world.

Furthermore, the scope of this research is restricted to quality-related issues given that the effect of other project management parameters (such as time, cost and risk) are considered as separate areas of further research.

PROJECT QUALITY, PROJECT EXCELLENCE AND SUSTAINABILITY

The title of this book highlights project quality as the primary theme of the book, but as indicated previously, a sustainable outcome also leads to project excellence. In this section the scope and parameters of project quality and project excellence are explained.

The success criteria of a project are the metrics and longer-term (or sustainable) outcomes which determine the degree of a success achieved by a project. The success factors are the enabling conditions and processes embedded in the project

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culture to ensure the success criteria of a project. For example 'on time' completion is one of the success criteria and stakeholder management is a success factor.

The initial literature review (Westerveld, 2003; Wateridge, 2002) indicates that there has not been adequate research to define concepts that can link projects success criteria to success factors. The concepts of project quality and project excellence are intended to link success criteria with success factors. *Project quality* is the philosophy of the adherence of standards throughout the life cycle of a project. *Project excellence* is the delivery of sustainable outcomes to key stakeholders during the life cycle of the project, particularly during the final closure stage. Project quality relates to the success factors and enablers and Project excellence relates to the success criteria and outcomes of a project. It will be demonstrated later that the dimensions of project quality are necessary but not sufficient contributors to project excellence.

It is also important to understand the span of project excellence and its interface with operational excellence. The life cycle of a project typically goes through four stages: initiation, design, execution, finalisation and follow-up (Turner, 1999). It is important to note, as shown in Figure I.1, that the span of project excellence follows the project life cycle although its effectiveness is enhanced at the final stage. Following the closure when the project is managed as operations by the operations team, it is the role of operational excellence to ensure, improve and sustain the outcomes delivered by the project beyond the project life cycle.

Another important and relevant area of understanding is sustainability in the context of managing quality in projects. I aim to examine key aspects of managing quality in major projects and to identify the major elements that contribute towards

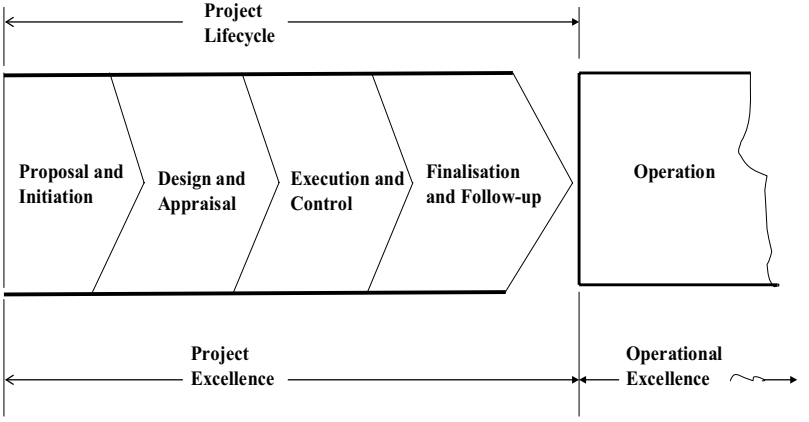


Figure I.1 The span of project excellence

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making the outcome of such projects sustainable. Sustainability, in a general sense, is the capacity to maintain a certain process or state indefinitely and in the context of preserving the environment, it relates to meeting 'the needs of the present without compromising the ability of future generations to meet their own needs' (United Nations, 1987). The sustainability of project outcomes, in the context of this research, is not environmental, it is the longer-lasting stability of the deliverables and processes derived from the PDCA cycle (Deming, 1986), as illustrated in Figure I.2.

An empirical study in operational management by Gallear et al (2000) supports the proposition that the longer-lasting outcomes of quality initiatives such as TQM are 'dependent on the integration of its underlying concepts with the normal business processes'. The sustainability of project outcomes, in the context of this research, is comparable to the stability of a process. Stability involves achieving consistent and, ultimately, higher process yields defined by a metric or a set point through the application of an improvement methodology and continuous review. The stability is ensured by minimising the variation of the set point. The approach of the plan, do, check, act cycle (PDCA) (Deming, 1986) enables both temporary and longer-term corrections. The temporary action is aimed at fixing the problem and the permanent corrective action consists of investigating and eliminating the

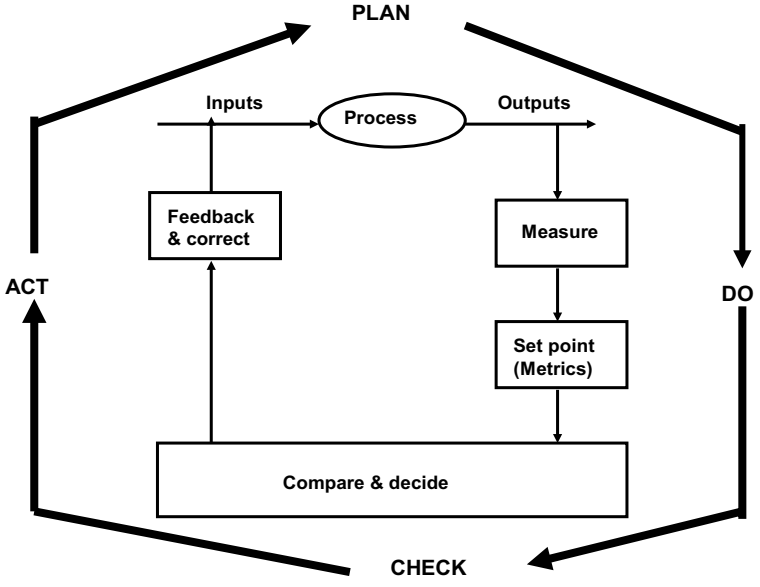


Figure I.2 A model showing a process towards a longer-lasing outcome in project management (developed from Deming, 1986)

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root causes and thus targets the sustainability of the improved process. Thus, in the context of project management and the scope of this book, a sustainable project is expected to deliver long-lasting outcomes of acceptable quality performance criteria (Mengel, 2008). When a project achieves defined criteria of sustainability (Edum-Fotwe and Price, 2009) in the context of this research, a project achieves project excellence (Abidin and Pasquire, 2007). This research will attempt to address these criteria of sustainability.

The above explanation of sustainable outcome in the context of major projects is also in agreement with the following definition of sustained success in British Standards. 'The sustained success of an organization is the result of its ability to achieve and maintain its objectives in the long-term. The achievement of sustained success for any organization is a complex and demanding challenge in an ever-changing environment' (BS EN ISO 9004: 2009).

THE PRACTICAL SIGNIFICANCE OF THIS BOOK

Phillips and Pugh (2000) distinguish research from what they call gathering of facts, and suggest that the significance of a research project should be demonstrated by practical and theoretical outcomes in seeking to explain a phenomenon and to analyse relationships. This section addresses the practical significance of this book.

The rationale behind selecting the topic of project quality in this book was partly influenced by the author's association with the Major Projects Association and thus has a strong practical significance. The theme is highly topical in view of the chequered history of a number of major projects in the UK, including the Dome Project, the Channel Tunnel, the new Wembley Stadium, Heathrow Terminal 5 and the expectations of outcomes from the London Olympic Games 2012 and the Crossrail project. A recent paper by Kaplan and Norton (2004) has concluded that measuring the value of sustainable criteria including intangible assets, although difficult to measure, is the Holy Grail of organisational culture, and these are worth far more to many companies in their longer-term performance than their immediate tangible results. This conclusion encourages the study on the role of organisation culture in the longer-term outcome of projects.

This book aims to be practically valuable to the furtherance of research-based operational information within the major projects sector, the critical evaluation of quality-related thinking on current project management practice and the objective application of this in a challenging sector of the economy. It also aims to develop a practical tool to assess and monitor the constructs contributing to project excellence.

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The research in this book is predominantly an applied research but it also contains its theoretical significance. The primary theoretical significance of this research is to reduce the aforementioned gap in academic research in defining the dimensions of project quality, and identifying the tools and processes to sustain longer-term outcomes leading to project excellence. From a theoretical point of view, the topic is important for several areas of academic study. First, a critical review of the literature relevant to research questions will develop propositions concerning the definitions and applications of quality and excellence in project management. A detailed review of the publications on research methodology will also bolster the appropriate mixed methodology for conducting research on social science topics such as project quality and excellence.

The theoretical focus of this research is on two dependable variables as units of analysis, viz, project quality and project excellence. The study aims to provide a theoretical underpining to link project quality relating to success factors and enablers and project excellence relating to the success criteria and outcomes of a project.

SUMMARY

This chapter serves as an introduction to the book and a broad understanding of the nature and importance of project quality. A preliminary literature review included in this chapter identifies a gap in the definition and application of project quality leading to some high-profile project failures. The analysis in this chapter correlates the poor outcomes of projects with the limited perspective that seems to be given to project quality. The commentary in this chapter also explains how the text and structure of this book will fill this gap and differ from the comparable publications on project quality.

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